ASEN 3802 -- Lab 3 -- Aerodynamics Lab

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Summary:

-- PART 1-- Task 1: models a 4 digit NACA airfoil both numerically and with an equi-angular representation. Task 2: Uses vortex panel code provided to find the coefficient of lift for different NACA airfoil cases. Also, we will analyze where the coefficient of lift converges based on the number of panels used, N. Task 3: Using thin airfoil theory equations, the effects of airfoil thickness on overall lift will be evaluated. -- PART 2 -- Task 1: Created Pradtl's lifting line theory function and recreated Anderson 5.20 plot to validate

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Last Revised: 4/14/2026

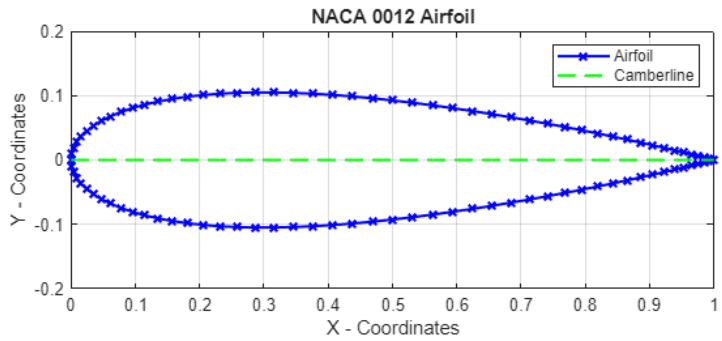
Housekeeping

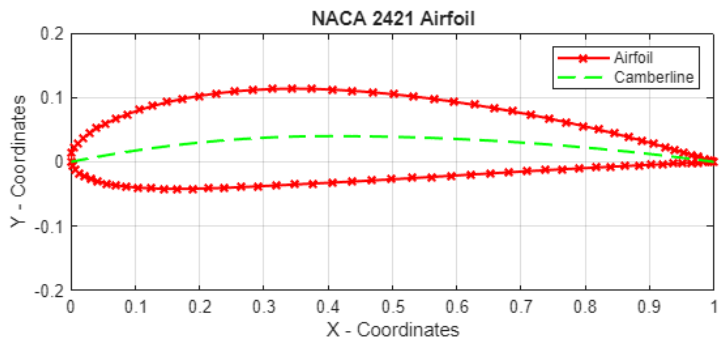
```
clc
clear
close all
```

Task 1 -- Plot of NACA 0021 and NACA 2421 using 50 panels

```
% Define Necessary parameters
```

```
% NACA 0021
m = 0/100;
p = 0/10;
t = 21/100;

% NACA 2421
m2 = 4/100;
p2 = 4/10;
t2 = 15/100;

c = 1;
N = 50;

% Call Airfoil Distribution function for each case
[x_b, y_b, x_c, y_c] =  NACA_Airfoils(m,p,t,c,N);
[x_b2, y_b2, x_c2, y_c2] =  NACA_Airfoils(m2,p2,t2,c,N);

% Plot both airfoil distributions
figure();
subplot(2, 1, 1);
plot(x_b, y_b, "-xb", "LineWidth",1.5)
hold on
plot(x_c, y_c, "--g", "LineWidth",1.5)
axis equal % implement correcting dimensions so that MATLAB does not stretch
the y-axis
xlim([0 c]) % set correct graph dimensions
ylim([-0.2*c 0.2*c])
grid on
xlabel('X - Coordinates');
ylabel('Y - Coordinates');
title("NACA 0012 Airfoil")
legend('Airfoil', 'Camberline')

subplot(2, 1, 2);
plot(x_b2, y_b2, "-xr", "LineWidth",1.5)
hold on;
plot(x_c2, y_c2, "--g", "LineWidth",1.5)
axis equal % implement correcting dimensions so that MATLAB does not stretch
the y-axis
xlim([0 c]) % set correct graph dimensions
ylim([-0.2*c 0.2*c])
grid on
xlabel('X - Coordinates');
ylabel('Y - Coordinates');
title("NACA 2421 Airfoil");
legend('Airfoil', 'Camberline')

print('task1', '-dpng', '-r500');

clc
clear
```



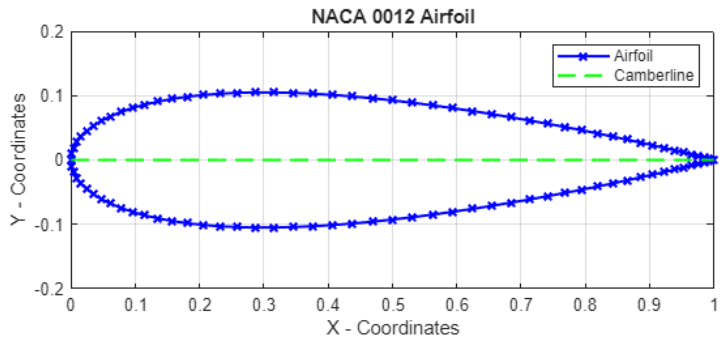


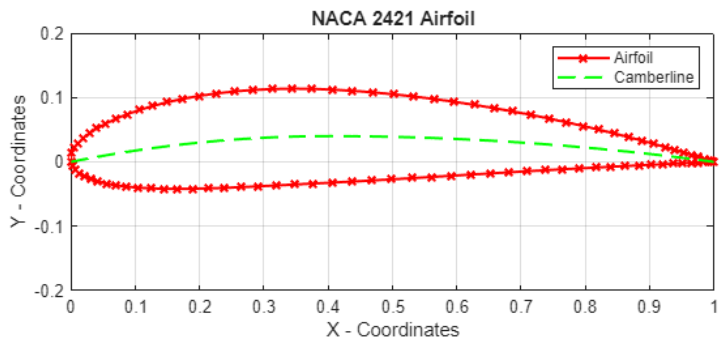
Task 2 --

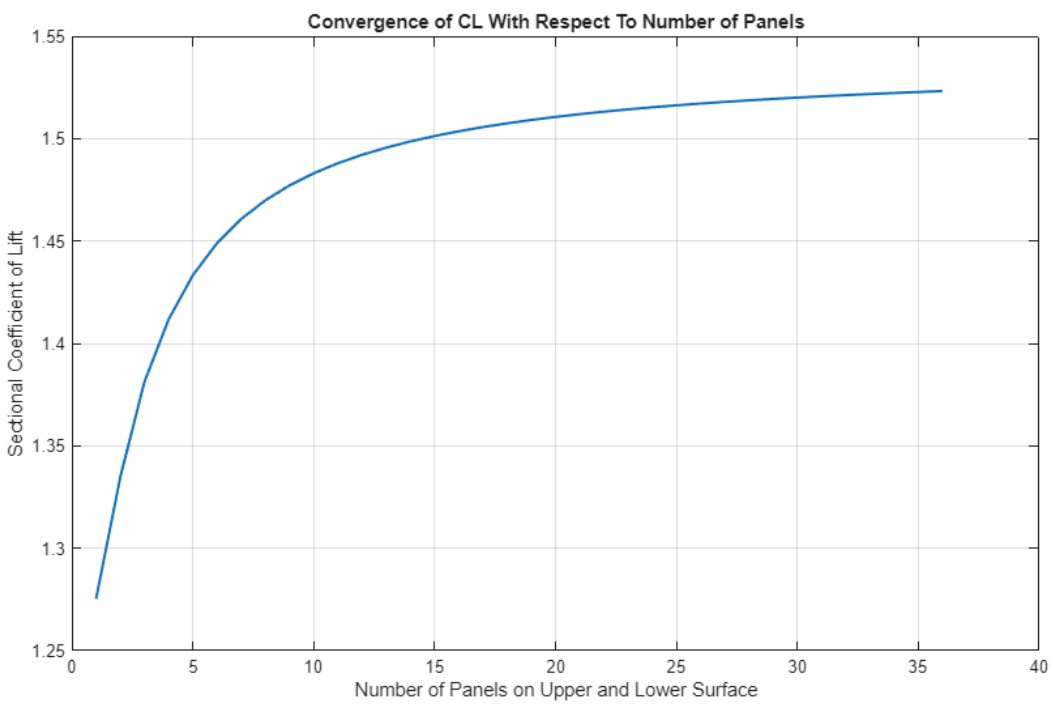
Define test variables:

```
i = 1;
m = 0/100;
p = 0/10;
t = 21/100;
c = 1;
CL_old = 0;
error = 1.5;
N_exact = 500;

alpha = 12; % [deg]
v_inf = 50; % [m/s]
Numerical_Plot = 1; % variable to turn on numerical plots for vortex panel
method

% Call numerical function to get point distribution
[x_b, y_b] =  NACA_Airfoils(m,p,t,c,N_exact);


% Call vortex panel method code to get output plot
[CL_accurate] = Vortex_Panel(x_b, y_b, v_inf, alpha);

N = 2;

% Loop through different vortex panel cases with incrementing N to determine
most ideal amount of
```

```
% panels to keep error under 1%.
while error > 1

alpha = 12; % [deg]
v_inf = 50; % [m/s]
Numerical_Plot = 1; % variable to turn on numerical plots for vortex panel
method

% Call numerical function to get point distribution
[x_b, y_b] =  NACA_Airfoils(m,p,t,c,N);


% Call vortex panel method code to get output plot
[CL(i)] = Vortex_Panel(x_b, y_b, v_inf, alpha);

% error = abs((CL(i)- CL_accurate)/(CL_accurate)) * 100;
error = abs((CL_accurate- CL(i))/(CL(i))) * 100;

% Assign new CL_value for the same error case
if error > 1
    N = N+1;
    CL_old = CL(i);
    i = i + 1;
end
end

N_vec = 1:length(CL);
disp(['Number of total panels: ', num2str(2*(length(N_vec)))])

% Plot the Number of panels versus the changing/converging CL value
figure();
plot(N_vec, CL,"LineWidth",1.5);
xlabel('Number of Panels on Upper and Lower Surface');
ylabel('Sectional Coefficient of Lift');
title('Convergence of CL With Respect To Number of Panels')
grid on

% Output results to a table
Results_Table = table( ...
    N_exact, CL_accurate, ...
    N_vec(end), CL(end), error(end), ...
    'VariableNames', {'N_exact', 'CL_exact', 'N_predicted', 'CL_predicted',
'Percent_Error'});
disp(Results_Table)
```







Task 3

```
% Call thin airfoil theory function for each different airfoil case -->
Note: for each case, alpha
```

```
% varies from -8 degrees to 8 degrees

% NACA 0006 case
m = 0/100;
p = 0/10;
t = 6/100;
c = 1;
N_req = N_vec(end); % vortex panels required is the last value of the
converging/increasing N vector

% Call thin airfoil theory function to determine conditions under thin
airfoil theory assumptions
[alpha_L0_thin_0006, alpha_L0_vort_0006, cl_thin_0006, CL_vort_0006,
slope_thin_0006, slope_vort_0006] = Thin_Airfoil_Theory(m, p, t, c, N_req,
alpha, v_inf);

% NACA 0012 case
m = 0/100;
p = 0/10;
t = 12/100;
c = 1;
N_req = N_vec(end); % vortex panels required is the last value of the
converging/increasing N vector

% Call thin airfoil theory function to determine conditions under thin
airfoil theory assumptions
[alpha_L0_thin_0012, alpha_L0_vort_0012, cl_thin_0012, CL_vort_0012,
slope_thin_0012, slope_vort_0012] = Thin_Airfoil_Theory(m, p, t, c, N_req,
alpha, v_inf);


% NACA 0018 case
m = 0/100;
p = 0/10;
t = 18/100;
c = 1;
N_req = N_vec(end); % vortex panels required is the last value of the
converging/increasing N vector

% Call thin airfoil theory function to determine conditions under thin
airfoil theory assumptions
[alpha_L0_thin_0018, alpha_L0_vort_0018, cl_thin_0018, CL_vort_0018,
slope_thin_0018, slope_vort_0018] = Thin_Airfoil_Theory(m, p, t, c, N_req,
alpha, v_inf);


% Read in experimental data
% experimental data from digitizer
data1 = load("NACA0006.mat");
data1 = data1.data;
AoA_exp0006 = data1(:,1);
CL_exp0006 = data1(1,:);

% Find the measured angle of attack value (i.e AoA_exp0006) that occurs when
```

```
the CL values are
% closest to 0 (for zero lift angle of attack)
[~, alpha_index_0006] = min(abs(CL_exp0006)); % finds index where CL is
closest to 0
AoA_L0_exp0006 = AoA_exp0006(alpha_index_0006); % Use this value for table
comparison


data2 = load("NACA0012.mat");
data2 = data2.data;
AoA_exp0012 = data2(:,1);
CL_exp0012 = data2(1,:);

% Find the measured angle of attack value (i.e AoA_exp0006) that occurs when
the CL values crosses 0
[~, alpha_index_0012] = min(abs(CL_exp0012)); % finds index where CL is
closest to 0
AoA_L0_exp0012 = AoA_exp0012(alpha_index_0012); % Use this value for table
comparison

data3 = load("NACA2412.mat");
data3 = data3.data;
AoA_exp2412 = data3(:,1);
CL_exp2412 = data3(1,:);

data4 = load("NACA4412.mat");
data4 = data4.data;
AoA_exp4412 = data4(:,1);
CL_exp4412 = data4(1,:);


% Create output tables
% Output table comparing zero lift angle of attack for each case

% NACA 0006 Airfoil
Results_Table = table( ...
    alpha_L0_vort_0006, alpha_L0_thin_0006, AoA_L0_exp0006, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0012 Airfoil
Results_Table = table( ...
    alpha_L0_vort_0012, alpha_L0_thin_0012, AoA_L0_exp0012, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0018 Airfoil -- Note experimental data does not exist for this
configuration
Results_Table = table( ...
    alpha_L0_vort_0018, alpha_L0_thin_0018, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory'});
disp(Results_Table)
```

```
% Output table comparing lift curve slope for each case

% NACA 0006 Airfoil
Results_Table = table( ...
    slope_vort_0006, slope_thin_0006, AoA_exp0006(1), ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0012 Airfoil
Results_Table = table( ...
    slope_vort_0012, slope_thin_0012, AoA_exp0012(1), ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0018 Airfoil -- Note experimental data does not exist for this
configuration
Results_Table = table( ...
    slope_vort_0018, slope_thin_0018, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory'});
disp(Results_Table)
```

---------- PART 2 ----------

```
% example inputs to test PLLT function
a0_t = 2*pi;
a0_r = 2*pi;
aero_t = 8;
aero_r = 8;
geo_t = 15;
geo_r = 15;
N = 100;
b = 1;
% define specified AR and taper ratio values to replicate Anderson figure
5.20
AR_vec = [4 6 8 10];
taper_ratio = linspace(0.01,1,100);
% Preallocate delta for plot
delta = zeros(length(AR_vec), length(taper_ratio));

for k = 1:length(AR_vec)
    AR = AR_vec(k);
for i = 1:length(taper_ratio) % compute geometric inputs based on AR and
taper ratio
    c_r = (2*b)./ (AR * (1+taper_ratio(i)));
    c_t = taper_ratio(i) * c_r;
    [e(k,i),c_L,c_Di] =
PLLT(b,a0_t,a0_r,c_t,c_r,aero_t,aero_r,geo_t,geo_r,N);
    delta(k,i) = (1/e(k,i)) - 1;
end
```

```
end
figure; hold on; grid on;
% Anderson 5.20 plot
plot(taper_ratio, delta(1,:), 'LineWidth', 2);
plot(taper_ratio, delta(2,:), 'LineWidth', 2);
plot(taper_ratio, delta(3,:), 'LineWidth', 2);
plot(taper_ratio, delta(4,:), 'LineWidth', 2);
xlabel('Taper Ratio (c_t/c_r)')
ylabel('\delta')
title('Effect of Aspect Ratio and Taper Ratio on \delta')
legend('AR = 4','AR = 6','AR = 8','AR = 10','Location','best')
```

User Defined Functions

1.) NACA Airfoil point distributions and plotting

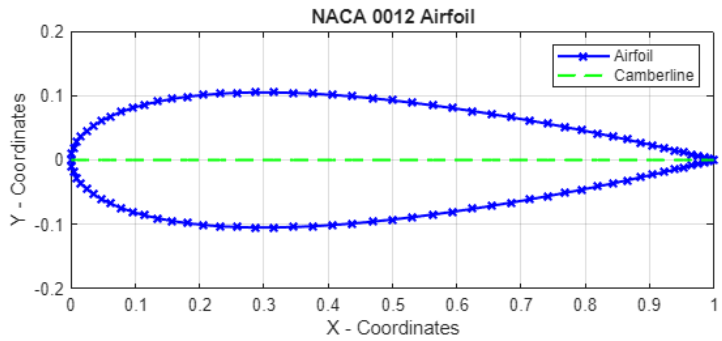
```
function [x_b,y_b,x_camber,y_camber] =  NACA_Airfoils(m,p,t,c,N)

% Goal/Purpose: Modeling a 4 digit NACA airfoil representation numerically
by converting 4 digit
% inputs into relevant variables and finding discrete x and y coordinates
for the top and bottom of
% the airfoil using numerically derived equations

% Inputs:
% 1.) m = max airfoil camber. This is the 1st digit (leftmost) of the 4
digit NACA airfoil
% representation and represents the max camber of the chord in percentage of
the chord.
% 2.) p = location of max camber. This is the 2nd digit from the left of the
4 digit NACA airfoil
% representation and represents the location of the max camber (measured
from the leading edge) in
% 1/10 of the chord.
% 3.) t = max airfoil thickness. This is the last 2 digits of the 4 digit
NACA airfoil
% representation and represents the maximum thickness of the airfoil in
percent of the chord.
% 4.) c = total chord length of the airfoil
% 5.) N = # of employed panels for vortex panel method over each surface

% Outputs:
% 1.) x_b = vector containing the x-location of all airfoil boundary points
% 2.) y_b = vector containing the y-location of all airfoil boundary points
```

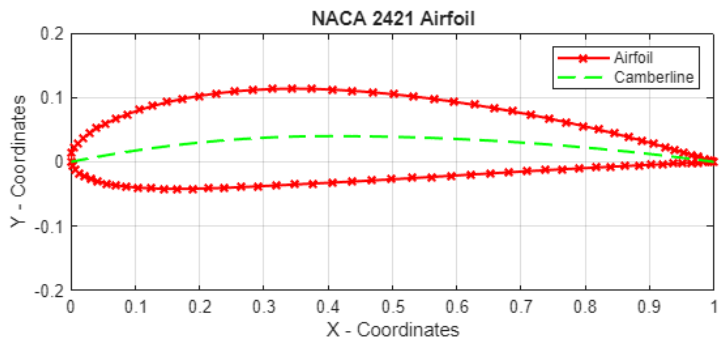
Define Thickness Distribution of Airfoil

```
for i = 1:N+1
    theta_i = (i-1)*pi/N;              % goes from 0 to pi
    x(i) = c/2 + c/2 * cos(theta_i); % x: c → 0
end
%disp(length(x));
```

```
y_t = ((t * c)/0.2) * ((0.2969 .* sqrt(x./c)) -  (0.1260*(x./c)) -
(0.3516 .* (x./c).^2) + (0.2843 .* (x./c).^3) - (0.1036 .* (x./c).^4)); %
numerical equation given
%disp(length(y_t));

% Define Camber Line Distribution

% preallocate variables:
y_c = zeros(size(x));
dy_c = zeros(size(x));
zeta = zeros(size(x));

x_u = zeros(size(x));
x_L = zeros(size(x));

y_u = zeros(size(x));
y_L = zeros(size(x));

% Define piecewise function using if statements

for i = 1:length(x)
    if m == 0 || p == 0

        y_c(i) = 0; dy_c(i) = 0; zeta(i) = 0;

    elseif x(i) < p*c

        y_c(i) = m * (x(i)/(p^2)) * ((2*p) - (x(i)/c)); % camber line
distribution

        dy_c(i) = ((2*m)/p) - ((2*m)/(c*p^2))*x(i); % derivative of camber
line distribution

        zeta(i) = atan(dy_c(i)); % compute zeta

    elseif x(i) >= p*c && x(i) <= c

        y_c(i) = m * ((c-x(i))/(1-p)^2) * (1 + (x(i)/c) - (2*p)); % camber
line distribution

        dy_c(i) = (2*p*m)/(1-p)^2 -((2*m)/(c*(1-p)^2))*x(i); % derivative of
camber line distribution

        zeta(i) = atan(dy_c(i)); % compute zeta

    else

        disp('Wrong x-value');

    end


    % Define X and Y locations over the upper and lower surfaces
```

```
        x_u(i) = x(i) - y_t(i)*sin(zeta(i)); % x-coordinates over the upper
surface
        x_L(i) = x(i) + y_t(i)*sin(zeta(i)); % x-coordinates over the lower
surface

        y_u(i) = y_c(i) + y_t(i)*cos(zeta(i)); % y-coordinates over the
upper surface
        y_L(i) = y_c(i) - y_t(i)*cos(zeta(i)); % y-coordinates over the
lower surface

    end


    % Note: we must reverse the order of all lower distributions to plot
clockwise
    % x_u_CW = flip(x_u);
    % y_u_CW = flip(y_u);
    % y_L_CW = -1*y_L_CW;
    % x_L_CW = -x_L_CW;


    % Group into final output variables

    % Note: output most be clockwise starting and ending at the trailing edge
    % gets rid of duplicate values at leading edge
    x_b = [x_L, flip(x_u(1:end-1))];
    y_b = [y_L, flip(y_u(1:end-1))];
    x_camber = fliplr(x);
    y_camber = fliplr(y_c);

end
```





2.) Vortex Panel Method Code

```
function [CL] = Vortex_Panel(XB,YB,VINF,ALPHA)

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Input:                            %
%                                   %
% XB   = Boundary Points x-location %
% YB   = Boundary Points y-location %
% VINF   = Free-stream Flow Speed   %
% ALPHA = AOA                       %
%                                   %
% Output:                           %
%                                   %
% CL = Sectional Lift Coefficient   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%
% Convert to Radians %
%%%%%%%%%%%%%%%%%%%%%%%%

ALPHA = ALPHA*pi/180;

%%%%%%%%%%%%%%%%%%%%%%%
% Compute the Chord %
%%%%%%%%%%%%%%%%%%%%%%%

CHORD = max(XB)-min(XB);
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Determine the Number of Panels %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

M = max(size(XB,1),size(XB,2))-1;
MP1 = M+1;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Intra-Panel Relationships:                                          %
%                                                                     %
% Determine the Control Points, Panel Sizes, and Panel Angles         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    IP1 = I+1;
    X(I) = 0.5*(XB(I)+XB(IP1));
    Y(I) = 0.5*(YB(I)+YB(IP1));
    S(I) = sqrt( (XB(IP1)-XB(I))^2 +( YB(IP1)-YB(I))^2 );
    THETA(I) = atan2( YB(IP1)-YB(I), XB(IP1)-XB(I) );
    SINE(I) = sin( THETA(I) );
    COSINE(I) = cos( THETA(I) );
    RHS(I) = sin( THETA(I)-ALPHA );
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Inter-Panel Relationships:              %
%                                         %
% Determine the Integrals between Panels  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    for J = 1:M
        if I == J
            CN1(I,J) = -1.0;
            CN2(I,J) = 1.0;
            CT1(I,J) = 0.5*pi;
            CT2(I,J) = 0.5*pi;
        else
            A = -(X(I)-XB(J))*COSINE(J) - (Y(I)-YB(J))*SINE(J);
            B = (X(I)-XB(J))^2 + (Y(I)-YB(J))^2;
            C = sin( THETA(I)-THETA(J) );
            D = cos( THETA(I)-THETA(J) );
            E = (X(I)-XB(J))*SINE(J) - (Y(I)-YB(J))*COSINE(J);
            F = log( 1.0 + S(J)*(S(J)+2*A)/B );
            G = atan2( E*S(J), B+A*S(J) );
            P = (X(I)-XB(J)) * sin( THETA(I) - 2*THETA(J) ) ...
              + (Y(I)-YB(J)) * cos( THETA(I) - 2*THETA(J) );
            Q = (X(I)-XB(J)) * cos( THETA(I) - 2*THETA(J) ) ...
              - (Y(I)-YB(J)) * sin( THETA(I) - 2*THETA(J) );
            CN2(I,J) = D + 0.5*Q*F/S(J) - (A*C+D*E)*G/S(J);
            CN1(I,J) = 0.5*D*F + C*G - CN2(I,J);
            CT2(I,J) = C + 0.5*P*F/S(J) + (A*D-C*E)*G/S(J);
            CT1(I,J) = 0.5*C*F - D*G - CT2(I,J);
```

```
        end
    end
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Inter-Panel Relationships:            %
%                                       %
% Determine the Influence Coefficients %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    AN(I,1) = CN1(I,1);
    AN(I,MP1) = CN2(I,M);
    AT(I,1) = CT1(I,1);
    AT(I,MP1) = CT2(I,M);
    for J = 2:M
        AN(I,J) = CN1(I,J) + CN2(I,J-1);
        AT(I,J) = CT1(I,J) + CT2(I,J-1);
    end
end
AN(MP1,1) = 1.0;
AN(MP1,MP1) = 1.0;
for J = 2:M
    AN(MP1,J) = 0.0;
end
RHS(MP1) = 0.0;

%%%%%%%%%%%%%%%%%%%%%%%%%
% Solve for the gammas %
%%%%%%%%%%%%%%%%%%%%%%%%%

GAMA = AN\RHS';

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Solve for Tangential Veloity and Coefficient of Pressure %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    V(I) = cos( THETA(I)-ALPHA );
    for J = 1:MP1
        V(I) = V(I) + AT(I,J)*GAMA(J);
    end
    CP(I) = 1.0 - V(I)^2;
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Solve for Sectional Coefficient of Lift %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

CIRCULATION = sum(S.*V);
CL = 2*CIRCULATION/CHORD;

end
```

```
Number of total panels: 72
    N_exact     CL_exact     N_predicted     CL_predicted     Percent_Error
    _______     ________     ___________     ____________     _____________

      500        1.538          36             1.5232           0.97338


    Vortex_Panel_Method     Thin_Airfoil_Theory     Experimental_Data
    ___________________     ___________________     _________________

        -1.1711e-16                 0                    -3.1111

    Vortex_Panel_Method     Thin_Airfoil_Theory     Experimental_Data
    ___________________     ___________________     _________________

        -2.0695e-16                 0                    -5.1556

    Vortex_Panel_Method     Thin_Airfoil_Theory
    ___________________     ___________________

         1.101e-18                  0

    Vortex_Panel_Method     Thin_Airfoil_Theory     Experimental_Data
    ___________________     ___________________     _________________

           6.519                  6.2832                -4.4444

    Vortex_Panel_Method     Thin_Airfoil_Theory     Experimental_Data
    ___________________     ___________________     _________________

          6.8324                  6.2832                -6.1333

    Vortex_Panel_Method     Thin_Airfoil_Theory
    ___________________     ___________________

          7.1493                  6.2832
```

3.) Thin Airfoil Theory Code

```
function [alpha_L0_thin, alpha_L0_vort, cl_thin, CL_vort, slope_thin,
slope_vort] = Thin_Airfoil_Theory(m, p, t, c, N_ideal, alpha,v_inf)

% Goal/Purpose: Find zero lift angle of attack using thin airfoil theory and
computing equation 4.61
% in anderson. Note that we already have dz/dx from previous derivation
input piecewise into our
% NACA_Airfoils function. Then, simply solve for theta from the equation: x
= (c/2)(1-cos(theta))
% and plug this in for theta and numerically integrate.

% Inputs:
% 2.) m = max airfoil camber. This is the 1st digit (leftmost) of the 4
```

```
digit NACA airfoil
% representation and represents the max camber of the chord in percentage of
the chord.
% 3.) p = location of max camber. This is the 2nd digit from the left of the
4 digit NACA airfoil
% representation and represents the location of the max camber (measured
from the leading edge) in
% 3.) t = max airfoil thickness. This is the last 2 digits of the 4 digit
NACA airfoil
% representation and represents the maximum thickness of the airfoil in
percent of the chord.
% 4.) c = chord length of given airfoil configuration
% 1/10 of the chord.
% 4.) N_ideal = ideal number of vortex panels to get within 1% error
% 5.) alpha = arbitrary input angle of attack value(s) in vector form

% Outputs:
% 1.) alpha_L0 = zero lift angle of attack using thin airfoil theory

alpha_vec = linspace(-8,8,N_ideal); %create an angle of attack vector
cooresponding to experimental trends
alpha_rad = (pi/180) .* alpha_vec;

% Define Thickness Distribution of Airfoil

% Equiangular distribution
for i = 1:N_ideal+1

    theta_i = (i-1)*pi/N_ideal;             % goes from 0 to pi

    x(i) = c/2 + c/2 * cos(theta_i); % x: c → 0

end

% y_t = ((t * c)/0.2) * ((0.2969 .* sqrt(x./c)) -  (0.1260*(x./c)) -
(0.3516 .* (x./c).^2) + (0.2843 .* (x./c).^3) - (0.1036 .* (x./c).^4)); %
numerical equation

% Define theta_0 in terms of x using chord line equation:

theta_0 =  acos( (1 - x.*(2/c)) );


% Define dz/dx for each part of the piecewise function bounds w.r.t length x


for i = 1:length(x)
    if m == 0 || p == 0 % for if there are any symmetric airfoil components

    dz_dx(i) = 0;

    elseif x(i) >= 0 && x(i) < p*c  % 1st part of piecewise function

    dz_dx(i) = ((2*m)/p) - ((2*m)/(c*p^2))*x(i); % derivative of camber line
```

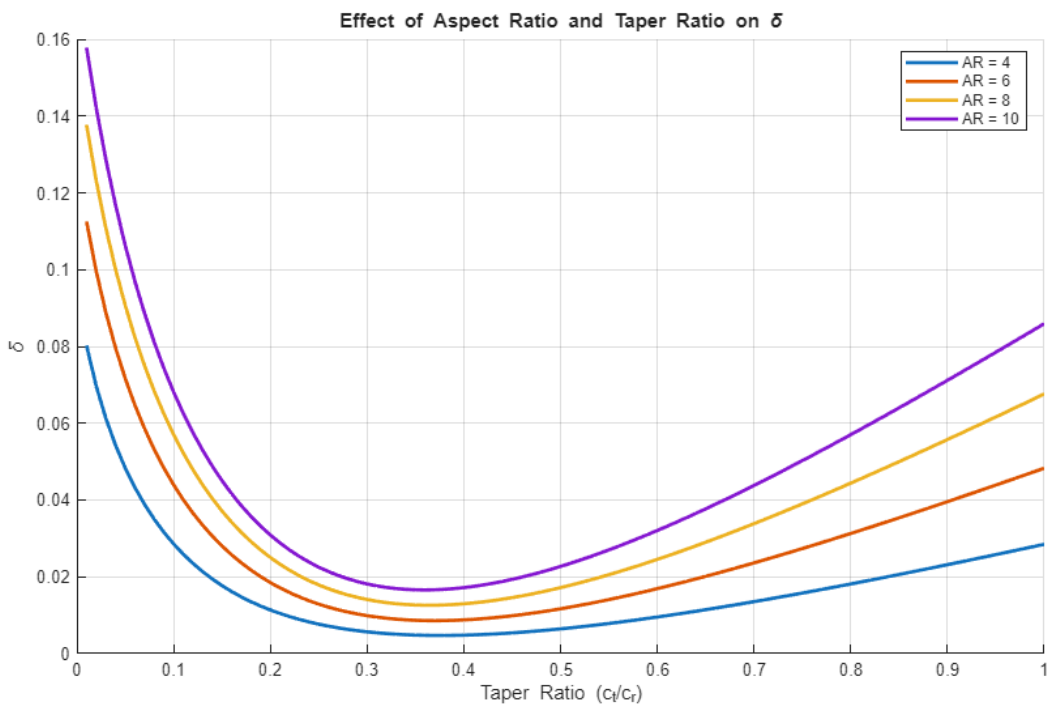
```
distribution


    elseif x(i) >= p*c && x(i) <= c

    dz_dx(i) = (2*p*m)/(1-p)^2 -((2*m)/(c*(1-p)^2))*x(i); % derivative of
camber line distribution

    else

    disp('Wrong x-value');

    end
end


% Define zero lift angle of attack with equation 4.61 and numerically
integrate

integrand = dz_dx.*(cos(theta_0) - 1);


alpha_L0_thin = (-1/pi) * trapz(theta_0, integrand); % computing zero lift
angle of attack in [rad] w.r.t theta using numerical integration

% loop through for a range of changing alpha values

[x_b, y_b] =  NACA_Airfoils(m,p,t,c,N_ideal);


% loop over all alpha vector values to get vector outputs of CL and cl for
each alpha case
for i = 1:length(alpha_vec)
CL_vort(i) = Vortex_Panel(x_b, y_b, v_inf, alpha_vec(i));
cl_thin(i) = (2*pi)*(alpha_rad(i) - alpha_L0_thin); % compute sectional lift
coefficient with given alpha value (note: alpha_L0 should be constant, only
alpha input indexes)
end

% calculate lift slope for both vortex and thin airfoil methods:

p_thin = polyfit(alpha_rad, cl_thin, 1); % ensure both alpha inputs and
cl_thin inputs have units of radians
slope_thin = p_thin(1);

p_vort = polyfit(alpha_rad, CL_vort,1);  % ensure both alpha inputs and
cl_thin inputs have units of radians
slope_vort = p_vort(1);
intercept_vort = p_vort(2);
alpha_L0_vort = -intercept_vort / slope_vort; % not sure if this is right.
should AoA L0 be a vector or one value?

end
```

4.) Prandtl Lifting Line Theory Code

Goal/Purpose: Analyze the performance of a finite wing by modeling a sheet of bound horseshoe vortex filaments. It solves the fundamental equation of Prandtl's Lifitng Line Theory for finite wings with thick airfoils to find the fourier coefficients, which can then be used to solve for various performace perameters such as e, CL, and CDi.

```
% Inputs:
% 1.) b = wing span (ft)
% 2.) a0_t = cross-sectional lift slope at the tips (per radian)
% 3.) a0_r = cross-sectional lift slope at the root (per radian)
% 4.) c_t = chord at the tips (in feet)
% 5.) c_r = chord at the root (in feet)
% 6.) aero_t = zero-lift angle of attack at the tips (in degrees)
% 7.) aero_r = zero-lift angle of attack at the root (in degrees)
% 8.) geo_t = geometric angle of attack at the tips (in degrees)
% 9.) geo_r = geometric angle of attack at the root (in degrees)
% 10.) N = number of odd terms to include in the series expansion for
circulation

% Outputs:
% 1.) e = span efficiency factor
% 2.) C_L = coefficient of lift
% 3.) C_Di = induced coefficient of drag

function [e,c_L,c_Di,delta] =
PLLT(b,a0_t,a0_r,c_t,c_r,aero_t,aero_r,geo_t,geo_r,N)
% convert angles of attack to radians
aero_t = aero_t * (pi/180);
aero_r = aero_r * (pi/180);
geo_t = geo_t * (pi/180);
geo_r = geo_r * (pi/180);

% satisfying the fundamental equation at N locations
thetas = (1:N) * pi/(2*N);
% using odd terms
n = 1:2:(2*N-1);
% non-dimensionalize distance along the wing
y = (b/2) .* cos(thetas);
y_normalized = abs(y)/(b/2);
% create linear spans along wing of inputs based on non-dimensionalization
c = c_r + (c_t - c_r) .* y_normalized;
a0 = a0_r + (a0_t - a0_r) .* y_normalized;
AoA_L0 = aero_r + (aero_t - aero_r) .* y_normalized;
AoA_geo = geo_r + (geo_t - geo_r) .* y_normalized;

% [b] = [A][x] where [x] contain fourier coefficients A1, A2 ....
b_vec = (AoA_geo - AoA_L0)';
% i corresponds to theta, j corresponds to N
% generate A matrix using the fundamental equation of PLLT
for i = 1:length(n)
    for j = 1:length(n)
        A_mat(i,j) = ((4*b)/(a0(i)*c(i))* sin(n(j)*thetas(i)) + (n(j) * ...
            (sin(n(j)*thetas(i))/sin(thetas(i)))));
```

```
    end
end

% x corresponds to Fourier coefficient column vector
x = A_mat\b_vec;
% solve for delta using solved Fourier Coefficients
delta = 0;
for i = 2:length(n)
    delta = delta + (n(i) * (x(i)/x(1))^2);
end
% span efficiency factor
e = 1 / (1 + delta);
% planform area
S = (b/2)*(c_r + c_t);
% Aspect ratio
AR = b^2/S;
% Solve for coefficient of lift using first fourier term
c_L = x(1)*pi*AR;

% compute Cdi by summing fourier coefficients
sum = 0;
for i = 1:length(n)
    sum = sum + (n(i) * x(i)^2) * (pi/2);
    c_Di = ((2*b^2)/S) * sum;
end

end
```



Published with MATLAB® R2025a